Zhe Li

Contact Information

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Summary

- 10+ years industrial and academic experiences/expertises in machine learning, deep learning from data collection system/protocol designing, data annotation/clean, model development, online/offline evaluation, deployment (quantilization/customized to hardware) on-device, the whole cycle of delivering billion-users level deep learning based features.
- Specialized in deep neural network optimization (Accelerating training [see publication section for reference, P4], Improving generalization ability [C6], Reducing model size [M1], and Searching optimal network structure [C1, P1]), strong mathematic skills, and deeply stay with the cutting edge research on and ability to prototype LLM and Gen-AI.
- Led to develop the first generation face recognition system for the world-class deep learning based feature Face ID unlock with mask and the second generation face recognition system for Face ID in iPhone 13 and later version.
- Currently as computer vision architecture for Apple Vision Pro Product, built up a broad understanding and overview from deep learning systems (hand tracking, eye tracking, Persona, body tracking, matting and others), software implementation, hardware availability and constraints (Computation (CPU, GPU, ANE), Memory, Camera sensor, ISP, SoC, Power) to the final product, evolved as an organized thinker and system level problem solver.

Working and Research Experiences

Member of Technical Staff, Patronus AI. Inc May 2026 - Present

- Leading the development of reinforcement learning environments for Computer Use, Tool Use, and Finance Domain (e.g., Bloomberg Terminal and SAP) for frontier AI labs.

Member of Technical Staff, Inflection AI. Inc Sep 2024 - May 2026

- Worked on large-scale distributed LLM training, inference and infra, resolving critical training failures to unblock the team and accelerate training experiments.
- Designed and implemented a batching system for LLM inference, improving GPU utilization by 3–4× and boosting throughput while lowering serving cost.
- Delivered core LLM inference features, including a structured output framework for schema-aligned, production-grade outputs.

Staff Machine Learning Engineer, Apple. Inc Sep 2022 - Apr 2024

- On-device computation resources (CPU, GUP, Apple Neural Engine (ANE)) allocation among different algorithms (hand tracking, eye tracking, persona, body tracking, matting and others).
- Data collection system designing (for hand tracking, body tracking, Persona).
- Proposed/Evaluated Camera number/position/spec, ISP streams, SoC requirement/definition.
- Involved/Investigated system-level solution for boosting frame rate for certain deep learning based algorithm.
- Investigated system level latency on hand tracking, eye tracking pipeline.

Senior Machine Learning Engineer, Apple. Inc Sep 2020 - Sep 2022

- Led to develop the 1st generation face recognition system for the world-class deep learning based feature FaceID unlock with mask.

- Responsible for the TrueDepth camera roadmap to FaceID feature.

Machine Learning Engineer, Apple. Inc Aug 2018 - Sep 2020

- Led to deliver the 2nd generation face recognition system for Face ID in iPhone 13.
- Responsible for Attention Awareness detection.
- Worked on Face Detection, Spoof Prevention.

Research Intern, Snap Research May 2017 - Nov 2017

- Reducing neural network size [M1]
- Searching optimal neural network structures[C1, P1]

Research Intern, Yahoo! Research May 2016 - Aug 2016

- Designed and implemented efficient optimization algorithms for solving SVM, Logistic Regression, Lasso.

Research Intern, General Motor May 2015 - Aug 2015

- Prototyped the real-time deep neural network based object detection system for autonomous driving.

Research Assistant, University of Iowa Aug 2013 - May 2018
Research Assistant, South Datota State University Aug 2011 - Jul 2013
Research Assistant, Xian Jiaotong University Sep 2010 - Jun 2011

Education

The University of Iowa, Iowa City, IA, USA
Ph.D., Computer Science Aug 2013 - May 2018

- Advisor: Tianbao Yang, Ph.D

South Dakota State University, Brookings, SD, USA
M.Sc., Electrical Engineering and Computing Science Sep 2011 - Jul 2013

Xi'an Jiaotong University, Xi'an, Shaanxi, China
M.Sc, Computer Science and Technology Aug 2010 - Jul 2011

Xi'an Jiaotong University, Xi'an, Shaanxi, China
B.Eng., Computer Science and Technology Aug 2006 - Jun 2010

Preprints

[P1] **Zhe Li**, Xuehan Xiong, Zhou Ren, Ning Zhang, Xiaoyu Wang, Tianbao Yang. "An Aggressive Genetic Programming Approach for Searching Neural Network Structure Under Computational Constraints"

[P2] **Zhe Li**, Xiaoyu Wang, Xutao Lv, Tianbao Yang. "SEP-Nets: Small and Effective Pattern Networks", arXiv preprint, arXiv:1703.01393. **Media Coverage**

[P3] Mingrui Liu, **Zhe Li**, Xiaoyu Wang, Tianbao Yang. "Adaptive Negative Curvature Descent with Application in Non-convex Optimization"

[P4] Tianbao Yang, Qihang Lin, **Zhe Li**. "Unified Convergence Analysis of Stochastic Momentum Methods for Convex and Non-convex Optimization", arXiv preprint, arXiv:1604.03257.

Refereed Conference Publications

[C1] Jian Ren, **Zhe Li**, Jianchao Yang, Ning Xu, Tianbao Yang, David J Foran "EIGEN: Ecologically-Inspired GENetic Approach for Neural Network Structure Searching from Scratch", In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2019

[C2] Dixian Zhu, **Zhe Li**, Xiaoyu Wang, Tianbao Yang "A Robust Zero-Sum Game Framework for Pool-based Active Learning"n Proceedings of the the 21st International Conference on Artificial Intelligence and Statistics (AISTATS), 2019

[C3] Tianbao Yang, **Zhe Li**, Lijun Zhang. "A Simple Analysis for Exp-concave Empirical Minimization with Arbitrary Convex Regularizer", In Proceedings of the the 21st International Conference on Artificial Intelligence and Statistics (AISTATS), 2018

[C4] Yichi Xiao, **Zhe Li**, Tianbao Yang, Lijun Zhang. "SVD-free Convex-Concave Approaches for Nuclear Norm Regularization", In Proceedings of the the 26th International Joint Conference on Artificial Intelligence(IJCAI). pages 3238 - 3244, 2017.

[C5] **Zhe Li**, Tianbao Yang, Lijun Zhang, Rong Jin. "A Two-stage Approach for Learning a Sparse Model with Sharp Excess Risk Analysis", In: Proceedings of the 30th AAAI Conference on Artificial Intelligence (AAAI'17), San Francisco, CA, 2017

[C6] **Zhe Li**, Boqing Gong, Tianbao Yang. "Improved Dropout for Shallow and Deep Learning", Nerual Information Processing System (NIPS) 29, Barcelona, Spain, Dec 2016.

[C7] **Zhe Li**, Tianbao Yang, Lijun Zhang, Rong Jin. "Fast and Accurate Refined Nyström based Kernel SVM", In: Proceedings of the 30th AAAI Conference on Artificial Intelligence (AAAI'16), Phoenix, AZ, 2016.

[C8] **Zhe Li**, Wei Wang, Sung Shin, Hyung D. Choi "Enhanced Roughness Index for Breast Cancer Benign/Malignant Measurement Using Gaussian Mixture Model", In Proc. ACM Research in Applied Computation Symposium (RACS), Oct. 2013.

[C9] Byung K. Jung, Wei Wang, **Zhe Li**, Seong H. Son, Jung Yeop Kim "A Sectorized Object Matching Approach for Breast Magnetic Resonance Image Similarity Study", In Proc. ACM Research in Applied Computation Symposium (RACS), Oct. 2012.

[C10] **Zhe Li**, Shin Sung, Soon I. Jeon, Seong H. Son, Jeong K. Pack "A New Histogram-based Breast Cancer Image Classifier Using Gaussian Mixture Model", In Proc. ACM Research in Applied Computation Symposium (RACS). Oct, 2012

Workshops

[M1] **Zhe Li**, Xiaoyu Wang, Xutao Lv, Tianbao Yang. "SEP-Nets: Small and Effective Pattern Networks". Beyond ISLVRC workshops 2017, Honolulu, Hi

[M2] **Zhe Li**, Tianbao Yang, Lijun Zhang, Rong Jin "A Two-stage Approach for Learning a Sparse Model with Sharp Excess Risk Analysis". Learning Fast From Easy Data workshop NIPS 2015, Montreal, Canada

Talks

- "SEP-Nets: Small and Effective Pattern Networks"
  Xi'an Jiaotong University, China — Aug, 2017
- "Deep Learning with Caffe"
  The University of Iowa, USA — Apr, 2017
- "Improved Dropout for Shallow and Deep Learning"
  Xi'an Jiaotong University, China — Dec, 2016
- "Open the Magic Box of Deep Learning for Image Classification"
  UIowa Graduate Research Symposium — Nov, 2016
- "Open the Magic Box of Deep Learning for Image Classification"
  Yahoo! Research — Aug, 2016
- "A Two-stage Approach for Learning a Sparse Model with Sharp Excess Risk Analysis"
  NIPS Learning Faster from Easy data Workshop, Canada — Dec, 2015
- "Nyström Based Kernel Classification for Big Data"
  1st Central State SIAM Conference, Rolla, MO,USA — Apr, 2015

Honors and Awards

- UIowa Graduate College Post-Comprehensive Research Award 2017
- Neural Information Processing Systems (NIPS) Travel Award 2016
- Outstanding Students Awards of Xi'an Jiaotong University, XJTU 2010
- BaoGang Outstanding Students Scholarship,XJTU 2009
- Outstanding Students Awards of Xi'an Jiaotong University, XJTU 2009
- National scholarship, XJTU 2008

References

Available upon requests